

NIEL ABHISHEK J DAVID

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PROFESSIONAL SUMMARY

Aspiring Site Reliability Engineer with a strong foundation in Java backend development, AWS cloud infrastructure (EC2, S3, Lambda), and monitoring automation backed by production-grade project experience. Proficient in Python, PySpark, Linux systems administration, CI/CD pipeline orchestration, and building self-healing automation workflows that improve system stability and operational efficiency. Passionate about high-performance, low-latency distributed systems—driving reliability, observability, and scalability across large-scale infrastructure.

EXPERIENCE

- ❖ **Intern | Indian Institute of Science (IISc), Bengaluru** Feb 2025 – May 2025
 - Improved backend microservice stability and application-specific performance by architecting FastAPI and WebSocket services for the MagicMirror² platform, enabling real-time, low-latency personalized UX at scale.
 - Integrated Firebase (Authentication, Firestore, Storage) as a secure, scalable backend data topology for user profiles, reducing manual state management overhead and ensuring reliable data access patterns.
 - Integrated **Gemini LLM** and **Face-Recognition** for personalized UX.
 - Delivered production-ready features across sprint cycles in an agile cross-functional team, applying performance tuning and reliability-first design practices to meet strict latency and uptime targets.
- ❖ **Graduate Apprentice | Volvo Group India Pvt.Ltd** March 2026 – Present
 - Engineered **Python** scripts to fix data entry in planning pipelines for aftermarket parts. This improved. Cut processing time by several hours each cycle.
 - Built. Automated reports using **SharePoint** and **Dataverse**. This sped up reporting. Showed key metrics to stakeholders.
 - Designed scalable, self-healing workflow infrastructure using Power Automate to streamline cross-system data flow, reducing manual intervention and contributing to enterprise-wide service reliability.
 - Performed root-cause analysis on production-adjacent pipeline failures, identifying data inconsistencies and driving process-level fixes — applying incident triage and post-mortem methodology in an agile environment.

PROJECTS

- ❖ **MagicMirror² | Digital Wardrobe**
Tech Stack: FastAPI, React.js, WebSockets, Gemini LLM, OpenCV, Face-Recognition, Firebase(Database)
 - Improved backend microservice stability and application-specific performance by architecting FastAPI and WebSocket services for the MagicMirror² platform, enabling real-time, low-latency personalized UX at scale.
 - Implemented secure, touchless authentication using facial recognition and pin-based verification ensuring reliable user identification and data access.
 - Integrated Gemini LLM with backend services to generate intelligent outfit recommendations based on wardrobe data, user preferences and contextual inputs.
- ❖ **Fashion Recommendation System**
Tech Stack: TensorFlow, EfficientNetB3, Flask, SQLAlchemy, React
 - Trained and deployed an EfficientNetB3-based deep learning model achieving 92% accuracy for body shape classification to enable personalized fashion recommendations.
 - Designed normalized SQL schemas using SQLAlchemy to store user measurements, preferences, and historical outfit data with efficient retrieval for personalization.
 - Developed Flask REST APIs to fetch user-specific data and deliver Amazon-sourced product recommendations, integrated seamlessly with React frontend.
- ❖ **Cloud-Based Attendance System**
Tech Stack: React.js, Node.js, Firebase Auth, Cloud Firestore, CI/CD, GitHub Actions
 - Achieved reliable, zero-downtime deployments by building end-to-end CI/CD automation pipelines with GitHub Actions, enforcing automated build validation and staged rollouts that mirror production SRE deployment workflows on cloud infrastructure.

- Built a cloud-native, scalable attendance management system on Firebase (Auth, Firestore) with role-based access control (RBAC), demonstrating ability to assess right-fit cloud topology for security, scalability, and availability requirements.
- Automated deployment workflows using CI/CD pipelines with GitHub Actions, ensuring reliable builds and faster feature delivery.

❖ **Firewall Monitoring Dashboard**

Tech Stack: Python, Flask, React, REST APIs

- Engineered a real-time network monitoring and alerting platform capable of detecting anomalous firewall rule violations and surfacing low-latency log streams, enabling operators to troubleshoot core network and performance issues within seconds of occurrence.
- Built RESTful backend APIs supporting CRUD operations for firewall rules and real-time log retrieval, reducing mean-time-to-detect (MTTD) network anomalies by centralising visibility across distributed rule sets.
- Designed the system topology for high-availability log ingestion and rule evaluation, applying the same observability principles (monitor high performance, alert on threshold breaches) required in production SRE environments.

❖ **Examination System**

Tech Stack: PHP, PHPUnit, MySQL, Selenium

- Developed a secure online examination platform supporting authentication, timed exams, and full CRUD operations for questions and results.
- Designed and validated MySQL database transactions to ensure data consistency and integrity during exam submissions.
- Containerized backend services using Docker to simplify deployment consistency across development and testing environments.

EDUCATION

❖ **B.Tech in Computer Science Engineering**

PES University, RR Campus, Bengaluru

Graduated: Aug 2025

CGPA– 7.73

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, Bash, JavaScript, PHP
- **OS & Systems:** Linux (RHEL/CentOS), Ubuntu, Bash, Shell Scripting
- **Backend Development:** FastAPI, Hibernate, Servlets, Spring Boot, Django
- **Database:** MySQL, MongoDB, Oracle SQL(PL/SQL)
- **Machine Learning Libraries:** TensorFlow, PyTorch, Pandas, NumPy, Scikit-Learn, spaCy, pyspark
- **Cloud & DevOps:** AWS (EC2, S3, Lambda)
- **Version Control & Containerization:** Docker, GitHub Actions CI/CD, Kubernetes, Git

CERTIFICATIONS

- Beginner's guide to Linux Kernel Development. ([Certificate](#))
- AWS educate Introduction to Cloud. ([Storage](#), [Compute](#), [Serverless](#))
- Jira Work Management Fundamentals Badge. ([Atlassian University](#))
- Java Full Stack & Generative AI

ACHIEVEMENTS

- Silver Medallist – India Skills Regional 2021 (Industrial Control)
- 6th Place – India Skills National 2022 (Industrial Control)
- Gold Medallist – Karnataka Skills State 2021 (Industrial Control)
- 8th Place – Kalpana Hackathon: Developed a disaster tracking and resource coordination system